

Understanding and Preparing the Dataset

When connecting YouTubeDataCleaned.csv with Category Names.csv, I retained only the 16 categories that appeared in the YouTube dataset. This prevents categories with no corresponding records from appearing in the category filters.

The same video can appear in multiple rows because a video may remain on the trending list for several days. Since views, likes, dislikes, and comments are cumulative measures, including all repeated rows would cause some videos to be counted multiple times and could distort sums and averages.

To address this issue, I first converted the Trending Date from a string to a date data type. I then created a calculated field to identify the latest trending date for each video, followed by a Boolean field that identifies whether each row represents that video's latest trending record. I used this filter when I needed each video to contribute only one observation.

Based on evaluator feedback, I also created FIXED level-of-detail calculations to identify the maximum recorded values for each video, such as {FIXED[Title]:MAX([Views])}, and created equivalent fields for likes, dislikes, and comment count. These maximum-per-video fields were then used in the relevant visualizations to ensure that repeated cumulative records did not inflate the results.

For analyses of trending frequency, I retained the repeated daily records because each row represents a video appearing on the trending list on a particular date. This allows trending appearances to be measured separately from cumulative performance metrics.

The location field appears to represent the creator's or video's associated location rather than the viewer's location. Therefore, the visualization "Trending Videos by Creator State" shows where trending videos were created or associated, not where their audiences were located. This interpretation should be confirmed against the dataset documentation.

I also created calculated fields for the percentage shares of likes and dislikes. These fields allow viewers to compare the proportion of positive and negative reactions within each category, with the two percentages totaling 100%.

The publication date cannot be used to measure category growth over YouTube's full history. Although some videos in the dataset were originally published many years earlier, the dataset only records videos that appeared on the trending list between November 2017 and March 2018. Older videos are included because they became trending during this period. Therefore, the analysis can only show changes in trending video performance within the available timeframe. A longer-term analysis would require additional data covering more trending dates.

I created an Engagement Rate calculated field using the following formula: Engagement Rate = $((\text{Likes} + \text{Comments}) / \text{Views}) \times 100$.

I created a Tableau story titled YouTube Trending Videos: A Category Analysis because the project examines differences in performance, engagement, trending frequency, channels, tags, and creator locations across YouTube video categories. The story contains five dashboards and 15 worksheets. The following sections describe the main findings and design choices for each dashboard.

Overall Design

I used YouTube Red as the main dashboard background color to create a consistent visual identity connected to the platform. I also used color-blind-friendly colors. Within dashboards with multiple charts, I used black chart backgrounds and white titles to separate each chart while maintaining strong visual contrast and readability.

I also kept formatting consistent across the story. Tooltips follow a similar structure, and colors have consistent meanings. For example, likes are represented in blue and dislikes in orange. Categories also retain the same colors across dashboards; for example, Music remains blue, and People & Blogs remains gray.

Each dashboard includes instructional text explaining how viewers can use legends and interactive selections.

Chart 1: Category Trends Over Time

Summary

This dashboard shows changes in average views across YouTube categories from November 2017 to March 2018. It allows viewers to examine questions such as:

- Which categories experienced the greatest increases or decreases in average views?
- Did the same categories consistently rank near the top?
- Did category performance remain stable or fluctuate throughout the period?

In this chart, we can see that Music consistently had the highest average views. In February 2018, it became second place. The first place was film & animation.

Design

I used a continuous line chart because the visualization focuses on change over time. Connecting the monthly observations with lines makes the direction and movement of average views easier to identify.

I created an interactive category legend that lets users highlight one category while the others turn gray. This makes it easier to follow a selected category across the timeline without completely removing the context provided by the others.

Dashboard 2: Category Performance Overview

Summary

This dashboard compares category performance across four measures:

- Average views
- Average comments
- Average engagement rate
- Trending appearances

Together, these charts demonstrate that strong performance in one measure does not necessarily indicate strong performance in another. For example, a category with high average views may not have the highest engagement rate or the most trending appearances.

The dashboard can help answer questions such as:

- Which category has the highest average views?
- Which category generates the most comments?
- Which category has the strongest engagement rates?
- Which category appears on the trending list most frequently?

In this dashboard, I discovered that although music had been the highest in average views, comments, but it was the second when it came to trending appearances. Entertainment appeared in trending days the most yet didn't have higher views or comments compared to other categories.

Design

I used four bar charts because they make it easy to compare differences among categories. Each chart is sorted by its corresponding measure, allowing viewers to identify high- and low-performing categories quickly.

The category legend is interactive. When a viewer selects a category, that category is highlighted across all four charts. This allows users to compare the selected category's relative performance across multiple measures.

For the trending-appearance chart, duplicate records were retained because each record represents one video that appeared on the trending list on a given date. For the other three charts, only the latest record for each video was used to prevent repeated cumulative values from distorting the averages.

Dashboard 3: Likes and Dislikes by Category

Summary

This dashboard compares average likes and average dislikes across YouTube categories. It also displays the percentage share of likes and dislikes within each category.

The dashboard allows viewers to examine questions such as:

- Which categories receive the highest average number of likes?
- Which categories receive the highest average number of dislikes?
- Which categories have the highest like or dislike shares?
- Do categories with many likes also receive many dislikes?

In this dashboard, we can see that the News & Politics category appears to have a higher dislike share than many other categories in the dataset.

Design

I used two vertical bar charts to display the average number of likes and dislikes by category. Placing these charts side by side makes it easier to compare their rankings and relative magnitudes.

For the percentage comparison, I used a 100% stacked horizontal bar chart. Each category's bar totals 100%, showing the proportion of likes versus dislikes. The horizontal format provides enough space for category labels and makes differences in reaction share easier to compare.

The category legend is interactive, allowing viewers to highlight a selected category across the dashboard. A separate legend identifies the like and dislike portions of the percentage chart.

Dashboard 4: Comment Count Relationships

Summary

This dashboard examines the relationships between comment count and four other measures:

- Likes
- Dislikes
- Views
- Engagement rate

The scatterplots suggest positive relationships between comment count and measures such as likes and views. In contrast, the visual relationship between engagement rate and comment count appears weaker.

The dashboard can help viewers explore questions such as:

- Do videos with more likes also tend to receive more comments?
- Is the comment count associated with the view count?
- How does engagement rate relate to comment activity?
- Do these relationships differ across categories?

In these scatterplots, I discovered that, compared to the other plots, engagement rate vs comment count showed no relationship, unlike like vs comment count or view vs comment count, which showed a positive relationship.

These visual patterns indicate association, not causation. A stronger statistical conclusion would require measures such as correlation coefficients or additional modeling.

Design

Because this dashboard focuses on relationships between numerical variables, I used four scatterplots. Each point represents a video, and the category is encoded by color. Viewers can use the category legend to highlight a specific category across all four charts.

I excluded the video “YouTube Rewind: The Shape of 2017” because it was an extreme outlier that compressed the remaining data points and made the overall patterns difficult to examine. Removing the outlier improved the visibility of relationships across most videos. However, this exclusion should be clearly disclosed because the video remains a valid observation.

Dashboard 5: Category Channels, Tags, and Locations

Summary

This interactive dashboard allows viewers to select a category and examine its associated channels, tags, and creator locations.

The dashboard shows:

- Channels with the greatest number of unique trending videos
- Tags with the greatest number of trending appearances
- States associated with the greatest number of unique trending videos

When no category is selected, the charts display results across the full dataset. When a category is selected, all charts update to show the channels, tags, and creator locations associated with that category.

The dashboard can help answer questions such as:

- Which channels produced the most unique trending videos?

- Which tags appeared most frequently among trending records?
- Which states were associated with the greatest number of trending videos?
- How do these patterns change when a specific category is selected?

In this dashboard, with no filters applied, we can see right away that “The Tonight Show Starring Jimmy Fallon” had the most unique trending videos. And most of the unique trending videos were associated with California. And the top tags were “funny,” followed by “comedy.”

Design

I used horizontal bar charts for channels and tags because both fields contain many categories and potentially long labels. The horizontal layout improves readability, and sorting the bars in descending order helps viewers quickly identify the highest-ranking channels and tags.

For the channel chart, each video is counted only once, even if it remained trending for several days. Therefore, the measure represents the number of unique trending videos produced by each channel.

For the tag chart, repeated appearances are retained so the visualization represents how frequently each tag appeared across trending records.

I used a filled map for creator locations because geographic data is more intuitive when displayed spatially. Darker shading represents a greater number of unique trending videos associated with a state. The map reflects the locations connected to creators or videos, not the geographic distribution of viewers.

A category filter controls the dashboard, allowing users to explore how channels, tags, and creator locations differ among selected YouTube categories.